

Chapter 2

Conditional Probability

2.1 The Definition of Conditional Probability

Commentary

It is useful to stress the point raised in the note on page 59. That is, conditional probabilities behave just like probabilities. This will come up again in Sec. 3.6 where conditional distributions are introduced.

This section ends with an extended example called “The Game of Craps”. This example helps to reinforce a subtle line of reasoning about conditional probability that was introduced in Example 2.1.5. In particular, it uses the idea that conditional probabilities given an event B can be calculated as if we knew ahead of time that B had to occur.

Solutions to Exercises

1. If $A \subset B$, then $A \cap B = A$ and $\Pr(A \cap B) = \Pr(A)$. So $\Pr(A|B) = \Pr(A)/\Pr(B)$.
2. Since $A \cap B = \emptyset$, it follows that $\Pr(A \cap B) = 0$. Therefore, $\Pr(A | B) = 0$.
3. Since $A \cap S = A$ and $\Pr(S) = 1$, it follows that $\Pr(A | S) = \Pr(A)$.
4. Let A_i stand for the event that the shopper purchases brand A on his i th purchase, for $i = 1, 2, \dots$. Similarly, let B_i be the event that he purchases brand B on the i th purchase. Then

$$\begin{aligned}\Pr(A_1) &= \frac{1}{2}, \\ \Pr(A_2 | A_1) &= \frac{1}{3}, \\ \Pr(B_3 | A_1 \cap A_2) &= \frac{2}{3}, \\ \Pr(B_4 | A_1 \cap A_2 \cap B_3) &= \frac{1}{3}.\end{aligned}$$

The desired probability is the product of these four probabilities, namely $1/27$.

5. Let R_i be the event that a red ball is drawn on the i th draw, and let B_i be the event that a blue ball is drawn on the i th draw for $i = 1, \dots, 4$. Then

$$\Pr(R_1) = \frac{r}{r+b},$$

$$\begin{aligned}
\Pr(R_2 \mid R_1) &= \frac{r+k}{r+b+k}, \\
\Pr(R_3 \mid R_1 \cap R_2) &= \frac{r+2k}{r+b+2k}, \\
\Pr(B_4 \mid R_1 \cap R_2 \cap R_3) &= \frac{b}{r+b+3k}.
\end{aligned}$$

The desired probability is the product of these four probabilities, namely

$$\frac{r(r+k)(r+2k)b}{(r+b)(r+b+k)(r+b+2k)(r+b+3k)}.$$

6. This problem illustrates the importance of relying on the rules of conditional probability rather than on intuition to obtain the answer. Intuitively, but incorrectly, it might seem that since the observed side is green, and since the other side might be either red or green, the probability that it will be green is $1/2$. The correct analysis is as follows: Let A be the event that the selected card is green on both sides, and let B be the event that the observed side is green. Since each of the three cards is equally likely to be selected, $\Pr(A) = \Pr(A \cap B) = 1/3$. Also, $\Pr(B) = 1/2$. The desired probability is $\Pr(A \mid B) = \left(\frac{1}{3}\right) / \left(\frac{1}{2}\right) = \frac{2}{3}$.

7. We know that $\Pr(A) = 0.6$ and $\Pr(A \cap B) = 0.2$. Therefore, $\Pr(B \mid A) = \frac{0.2}{0.6} = \frac{1}{3}$.

8. In Exercise 2 in Sec. 1.10 it was found that $\Pr(A \cup B \cup C) = 0.85$. Since $\Pr(A) = 0.6$, it follows that $\Pr(A \mid A \cup B \cup C) = \frac{0.60}{0.85} = \frac{12}{17}$.

9. (a) If card A has been selected, each of the other four cards is equally likely to be the other selected card. Since three of these four cards are red, the required probability is $3/4$.
(b) We know, without being told, that at least one red card must be selected, so this information does not affect the probabilities of any events. We have

$$\Pr(\text{both cards red}) = \Pr(R_1) \Pr(R_2 \mid R_1) = \frac{4}{5} \cdot \frac{3}{4} = \frac{3}{5}.$$

10. As in the text, let π_0 stand for the probability that the sum on the first roll is either 7 or 11, and let π_i be the probability that the sum on the first roll is i for $i = 2, \dots, 12$. In this version of the game of craps, we have

$$\begin{aligned}
\pi_0 &= \frac{2}{9}, \\
\pi_4 &= \pi_{10} = \frac{3}{36} \cdot \frac{\frac{3}{36}}{\frac{3}{36} + \frac{6}{36} + \frac{2}{36}} = \frac{1}{44}, \\
\pi_5 &= \pi_9 = \frac{4}{36} \cdot \frac{\frac{4}{36}}{\frac{4}{36} + \frac{6}{36} + \frac{2}{36}} = \frac{1}{27}, \\
\pi_6 &= \pi_8 = \frac{5}{36} \cdot \frac{\frac{5}{36}}{\frac{5}{36} + \frac{6}{36} + \frac{2}{36}} = \frac{25}{468}.
\end{aligned}$$

The probability of winning, which is the sum of these probabilities, is 0.448.

11. This is the conditional version of Theorem 1.5.3. From the definition of conditional probability, we have

$$\begin{aligned}\Pr(A^c|B) &= \frac{\Pr(A^c \cap B)}{\Pr(B)}, \\ 1 - \Pr(A|B) &= 1 - \frac{\Pr(A \cap B)}{\Pr(B)}, \\ &= \frac{\Pr(B) - \Pr(A \cap B)}{\Pr(B)}.\end{aligned}\tag{S.2.1}$$

According to Theorem 1.5.6 (switching the names A and B), $\Pr(B) - \Pr(A \cap B) = \Pr(A^c \cap B)$. Combining this with (S.2.1) yields $1 - \Pr(A|B) = \Pr(A^c|B)$.

12. This is the conditional version of Theorem 1.5.7. Let $A_1 = A \cap D$ and $A_2 = B \cap D$. Then $A_1 \cup A_2 = (A \cup B) \cap D$ and $A_1 \cap A_2 = A \cap B \cap D$. Now apply Theorem 1.5.7 to determine $\Pr(A_1 \cup A_2)$.

$$\begin{aligned}\Pr([A \cup B] \cap D) &= \Pr(A_1 \cup A_2) = \Pr(A_1) + \Pr(A_2) - \Pr(A_1 \cap A_2) = \Pr(A \cap D) \\ &\quad + \Pr(B \cap D) - \Pr(A \cap B \cap D).\end{aligned}$$

Now, divide the extreme left and right ends of this string of equalities by $\Pr(D)$ to obtain

$$\begin{aligned}\Pr(A \cup B|D) &= \frac{\Pr([A \cup B] \cap D)}{\Pr(D)} = \frac{\Pr(A \cap D) + \Pr(B \cap D) - \Pr(A \cap B \cap D)}{\Pr(D)} \\ &= \Pr(A|D) + \Pr(B|D) - \Pr(A \cap B|D).\end{aligned}$$

13. Let A_1 denote the event that the selected coin has a head on each side, let A_2 denote the event that it has a tail on each side, let A_3 denote the event that it is fair, and let B denote the event that a head is obtained. Then

$$\begin{aligned}\Pr(A_1) &= \frac{3}{9}, \quad \Pr(A_2) = \frac{4}{9}, \quad \Pr(A_3) = \frac{2}{9}, \\ \Pr(B | A_1) &= 1, \quad \Pr(B | A_2) = 0, \quad \Pr(B | A_3) = \frac{1}{2}.\end{aligned}$$

Hence,

$$\Pr(B) = \sum_{i=1}^3 \Pr(A_i) \Pr(B | A_i) = \frac{4}{9}.$$

14. We partition the space of possibilities into three events B_1, B_2, B_3 as follows. Let B_1 be the event that the machine is in good working order. Let B_2 be the event that the machine is wearing down. Let B_3 be the event that it needs maintenance. We are told that $\Pr(B_1) = 0.8$ and $\Pr(B_2) = \Pr(B_3) = 0.1$. Let A be the event that a part is defective. We are asked to find $\Pr(A)$. We are told that $\Pr(A|B_1) = 0.02$, $\Pr(A|B_2) = 0.1$, and $\Pr(A|B_3) = 0.3$. The law of total probability allows us to compute $\Pr(A)$ as follows

$$\Pr(A) = \sum_{j=1}^3 \Pr(B_j) \Pr(A|B_j) = 0.8 \times 0.02 + 0.1 \times 0.1 + 0.1 \times 0.3 = 0.056.$$

15. The analysis is similar to that given in the previous exercise, and the probability is 0.47.

16. In the usual notation, we have

$$\begin{aligned}\Pr(B_2) &= \Pr(A_1 \cap B_2) + \Pr(B_1 \cap B_2) = \Pr(A_1) \Pr(B_2 \mid A_1) + \Pr(B_1) \Pr(B_2 \mid B_1) \\ &= \frac{1}{4} \cdot \frac{2}{3} + \frac{3}{4} \cdot \frac{1}{3} = \frac{5}{12}.\end{aligned}$$

17. Clearly, we must assume that $\Pr(B_j \cap C) > 0$ for all j , otherwise (2.1.5) is undefined. By applying the definition of conditional probability to each term, the right side of (2.1.5) can be rewritten as

$$\sum_{i=1}^k \frac{\Pr(B_j \cap C)}{\Pr(C)} \frac{\Pr(A \cap B_j \cap C)}{\Pr(B_j \cap C)} = \frac{1}{\Pr(C)} \sum_{i=1}^k \Pr(A \cap B_j \cap C).$$

According to the law of total probability, the last sum above is $\Pr(A \cap C)$, hence the ratio is $\Pr(A|C)$.

2.2 Independent Events

Commentary

Near the end of this section, we introduce conditionally independent events. This is a prelude to conditionally independent and conditionally i.i.d. random variables that are introduced in Sec. 3.7. Conditional independence has become more popular in statistical modeling with the introduction of latent-variable models and expert systems. Although these models are not introduced in this text, students who will encounter them in the future would do well to study conditional independence early and often.

Conditional independence is also useful for illustrating how learning data can change the distribution of an unknown value. The first examples of this come in Sec. 2.3 after Bayes' theorem. The assumption that a sample of random variables is conditionally i.i.d. given an unknown parameter is the analog in Bayesian inference to the assumption that the random sample is i.i.d. marginally. Instructors who are not going to cover Bayesian topics might wish to bypass this material, even though it can also be useful in its own right. If you decide to not discuss conditional independence, then there is some material later in the book that you might wish to bypass as well:

- Exercise 23 in this section.
- The discussion of conditionally independent events on pages 81–84 in Sec. 2.3.
- Exercises 12, 14 and 15 in Sec. 2.3.
- The discussion of conditionally independent random variables that starts on page 163.
- Exercises 13 and 14 in Sec. 3.7.
- Virtually all of the Bayesian material.

This section ends with an extended example called “The Collector’s Problem”. This example combines methods from Chapters 1 and 2 to solve an easily stated but challenging problem.